

SeaweedFS — Operator Manual

What this technology does

SeaweedFS stores the immutable/raw gateway-evidence objects that the verifier reopens later. PostgreSQL keeps metadata, hashes, references, leases, and verification state; SeaweedFS keeps the retained byte objects those records point to.

The commissioned object store is part of the evidence trust path. It is **not** scratch storage and it must not be cleared when telemetry tables are reset for a new research epoch.

Current commissioned topology

SeaweedFS OSS 4.41 is deployed on all three cloud hosts. Each Droplet is one SeaweedFS rack inside the same sgp1 data center.

```

ULC-01 10.104.0.2 -> metadata-etcd + SeaweedFS master/volume/filer/S3
ULC-02 10.104.0.4 -> metadata-etcd + SeaweedFS master/volume/filer/S3
ULC-03 10.104.0.8 -> metadata-etcd + SeaweedFS master/volume/filer/S3

metadata-etcd client/peer    12379 / 12380
master HTTP/gRPC            19333 / 19334
volume HTTP/gRPC            18082 / 18083
filer HTTP/gRPC              18888 / 18889
raw S3 backend                127.0.0.1:18333
S3 gRPC                      18334
TLS/create-only frontend    node VPC :18443
logical endpoint              evidence-objects.internal.lorawan.com:18443
bucket                       lorawan-evidence
placement                    010

```

Placement 010 means one additional copy in another rack in the same data center. Because each Droplet is modeled as a different rack, an acknowledged raw object has two host copies.

SeaweedFS uses a **separate** three-member metadata-etcd cluster on 12379/12380. Never point the filer at the Patroni DCS etcd on 2379/2380.

Current protected layout

```

/etc/lorawan-cloud/seaweedfs/release.env
/etc/lorawan-cloud/seaweedfs/host.env
/etc/lorawan-cloud/seaweedfs/filer.toml
/etc/lorawan-cloud/seaweedfs/s3.json
/etc/lorawan-pki/evidence-objectstore/
/srv/seaweedfs/metadata-etcd
/srv/seaweedfs/master
/srv/seaweedfs/volume

containers:
  seaweedfs-metadata-etcd
  seaweedfs

```

Do not print or copy live S3 credentials from s3.json.

Security and immutability boundary

The evidence services do not connect directly to the raw S3 listener. The raw S3 listener stays loopback-only on 127.0.0.1:18333.

Applications use the TLS endpoint on :18443, where HAProxy enforces:

GET / HEAD	allowed
PUT with exact If-None-Match: *	allowed
unconditional PUT	denied
POST	denied
DELETE	denied
other methods	denied

Ingest and collector identities can read/list/create evidence objects. The verifier is read/list only. No runtime identity receives SeaweedFS administration authority.

Step 1 — Check both containers on every host

Run on ULC-01, ULC-02, and ULC-03:

```
sudo docker ps --filter 'name=seaweedfs-metadata-etcd' --filter 'name=seaweedfs' --format 'table
{{.Names}}\t{{.Status}}\t{{.Image}}'

sudo docker inspect seaweedfs-metadata-etcd seaweedfs --format '{{.Name}} status={{.State.Status}}
restarts={{.RestartCount}}'
```

PASS means: both containers are running on every host and neither is in a restart loop.

A running process alone is not enough; continue through metadata quorum, listener, and object-store checks.

Step 2 — Verify the custom listener map

```
sudo ss -lntp | grep -E ':(12379|12380|19333|19334|18082|18083|18888|18889|18333|18334|18443)\b'
```

PASS means:

- metadata-etcd and Seaweed core ports are present on the intended private/loopback addresses;
- raw S3 :18333 is loopback-only;
- HAProxy :18443 is the service-facing TLS endpoint on the node VPC address.

Do not change the ports to generic Seaweed defaults. The custom allocation avoids existing service collisions.

Step 3 — Verify the separate metadata-etcd member

```
sudo docker exec -e ETCDCTL_API=3 seaweedfs-metadata-etcd etcdctl --endpoints=http://127.0.0.1:12379
endpoint health

sudo docker exec -e ETCDCTL_API=3 seaweedfs-metadata-etcd etcdctl --endpoints=http://127.0.0.1:12379 member
list -w table
```

PASS means: the local endpoint is healthy and the common cluster view contains the intended three members.

Do not repair this cluster with the Patroni etcd data directory, cluster token, or peer set.

Step 4 — Verify the effective replication arguments

```
sudo docker inspect seaweedfs --format '{{json .Config.Cmd}}' | tr ',' '\n' | grep -E
'defaultReplication|defaultReplicaPlacement|dataCenter|rack'
```

PASS means: the effective command still carries replication 010, the expected data center, and the node-specific rack identity.

A directory existing on two hosts is not equivalent to proven object-store replication.

Step 5 — Verify the TLS/create-only frontend

Inspect the certificate and perform a bounded server-auth TLS check with the commissioned CA:

```
openssl s_client -connect evidence-objects.internal.lorawan.com:18443 -servername evidence-
objects.internal.lorawan.com -verify_hostname evidence-objects.internal.lorawan.com -verify_return_error
-CAfile /etc/lorawan-pki/evidence-objectstore/ca.crt </dev/null 2>/dev/null | grep -E 'Verify return
code|subject=|issuer='
```

PASS means: hostname and chain validation succeed. Do not use -k or disable certificate verification.

The commissioned S9 create/idempotency/race/cross-host contract has already passed. Do **not** create new commissioning objects merely to prove a documentation change.

Step 6 — Check storage and recent logs

```
df -h /srv/seaweedfs
sudo du -sh /srv/seaweedfs/metadata-etcd /srv/seaweedfs/master /srv/seaweedfs/volume
sudo docker logs --since=15m --tail=120 seaweedfs-metadata-etcd
sudo docker logs --since=15m --tail=160 seaweedfs
```

Look for quorum loss, master/filer membership churn, volume errors, write failures, corruption warnings, OOM/restart events, or sustained disk pressure.

Do not delete retained evidence objects or commissioning fixtures to solve disk usage without an explicit retention/recovery decision.

Step 7 — Verify application-facing object-store readiness

The normal evidence-service configuration is:

```
EVIDENCE_OBJECTSTORE_BACKEND=s3
EVIDENCE_S3_ENDPOINT=https://evidence-objects.internal.lorawan.com:18443
EVIDENCE_S3_BUCKET=lorawan-evidence
EVIDENCE_S3_PREFIX=lorawan-gateway-evidence
```

Use the Gateway Evidence Services manual to prove ingest/collector/verifier /readyz state. Their readiness checks include the object-store dependency and are the preferred non-destructive application-level proof.

If all Seaweed containers are healthy but those services report object-store failure, inspect CA trust, HAProxy :18443, local name mapping, role credentials, and method policy before changing Seaweed data.

Safe configuration change procedure

1. Prove all three metadata-etcd members and all three Seaweed containers are healthy.
2. Record current image IDs and effective arguments.
3. Back up the exact configuration being changed.
4. Change one node at a time.
5. Validate the Compose/configuration model before recreate.
6. Preserve metadata quorum and a healthy raw-object copy throughout.
7. Verify local listeners and metadata membership.
8. Verify :18443 TLS.
9. Verify evidence-service readiness.
10. Re-run the destructive/empirical S9 object contract only when a storage change actually requires re-acceptance.

Do not restart all three storage members together for an ordinary configuration edit.

Backup and recovery rules

Replication is availability, not a complete backup policy. Preserve protected deployment/configuration, metadata-etcd recovery material using its reviewed snapshot procedure, retained raw evidence, PostgreSQL evidence metadata that references those objects, and the internal object-store CA/required service identities.

Do not copy a live etcd data directory as a substitute for a consistent snapshot. Do not restore PostgreSQL metadata without considering the matching retained raw objects.

Troubleshooting map

Symptom	First check
both containers down on one host	Docker/runtime, protected env/config, disk, image
metadata-etcd unhealthy	separate 12379/12380 quorum and peer reachability
Seaweed up but application S3 fails	:18443 HAProxy/TLS/name mapping/role credentials
raw S3 reachable on non-loopback	security regression; fix bind immediately
evidence object missing	object reference, replication/member health, retained object; do not fabricate replacement
DB metadata exists but verifier cannot read bytes	object-store reachability/hash/reference before verifier code
disk pressure	retained-object growth and volume capacity; do not delete evidence casually
only one host has object	replication/placement investigation before accepting new evidence writes

Completion checklist

- seaweedfs-metadata-etcd and seaweedfs are running on all three hosts;
- metadata-etcd shows the intended three-member cluster;
- custom port map is intact;
- raw S3 remains loopback-only;
- HAProxy :18443 passes TLS hostname verification;
- effective placement remains 010;
- no immediate storage/quorum/restart errors exist;
- evidence-service readiness passes;
- retained raw evidence and commissioning fixtures remain intact.

Detailed references

- [../../../../evidence-services/cloud/deploy/seaweedfs/README.md](#)
- [../../../../evidence-services/cloud/deploy/seaweedfs/01-live-commissioning.md](#)
- [../server/integrations/gateway-integrity/06-replicated-ha-deployment-journey.md](#)

Supporting runbook: README.md

This is the original project runbook. Check the current documentation baseline and actual host state before applying historical steps or changing a live system.

SeaweedFS HA Raw-Evidence Store

This directory freezes the self-hosted production candidate for gateway raw evidence. It replaces the earlier managed-S3 placeholder with SeaweedFS while preserving the Go services' generic S3 interface.

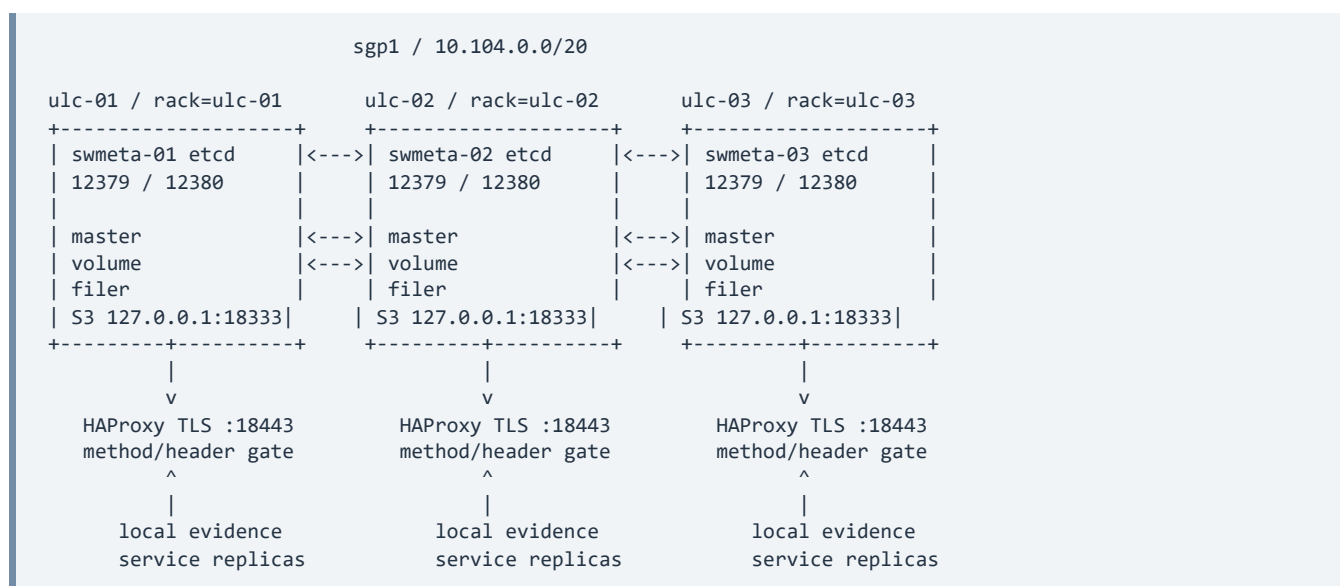
Selection: SeaweedFS OSS 4.41, live linux/amd64 image

chrislusf/seaweedfs:4.41@sha256:43b768cd62b00d132439cda881b93fd1adebf1b315e996e794087743821d771d,
commit de34a1a87. Dedicated filer metadata uses

quay.io/coreos/etcd:v3.5.15@sha256:0934690612905554eb61ddefb9faaaecb47c2f6931dbb453e694358092ee8990.

Releases before 4.30 are not acceptable because the S3 gateway had a fixed cross-bucket path-traversal vulnerability.

Topology



SeaweedFS placement is 010: one additional copy in a **different rack in the same data center**. Each physical Droplet is deliberately one rack, so an acknowledged raw object has two host copies. Do not replace this with 001; 001 means another server in the same rack and does not describe our host-failure model as clearly.

The separate `swmeta-* etcd` quorum exists only for SeaweedFS filer namespace metadata. It uses 12379/12380, separate data directories, and a separate cluster token. Never point the filer at the commissioned Patroni DCS etcd on 2379/2380.

Why the runtime S3 endpoint is stricter than SeaweedFS IAM

SeaweedFS's `write` action includes delete-capable S3 operations. Therefore IAM alone is **not** our immutability boundary.

The evidence services only require:

```
HEAD bucket/object readiness
GET object bytes
PUT object with If-None-Match: *
```

The host HAProxy frontend enforces exactly that surface:

GET / HEAD	allowed
PUT + If-None-Match: *	allowed
PUT without create-only condition	denied
POST	denied
DELETE	denied
all other methods	denied

This means a runtime S3 credential cannot use the commissioned endpoint to issue object deletion, multipart completion, bucket administration, ACL mutation, or unconditional overwrite. SeaweedFS's own conditional write then arbitrates two concurrent create attempts atomically. Filer WORM may be enabled later as defense in depth only after its exact behavior is empirically proven; correctness does not depend on assuming undocumented retention semantics.

The raw Seaweed S3 listener binds 127.0.0.1:18333, so application containers never connect to it directly. They connect to `https://evidence-objects.internal.lorawan.com:18443`, resolved to the local host VPC address. TLS is terminated by the already-managed host HAProxy with an internal-PKI certificate.

Resource policy

The three application servers have only 2 GiB RAM. The candidate therefore uses:

```
SeaweedFS all-in-one process: 256 MiB container cap, GOMEMLIMIT=220MiB
Seaweed metadata etcd:       96 MiB container cap
volume index:                 leveldb, not in-memory
S3 chunk cache:               disabled
volume size limit:           1 GiB
```

These are commissioning caps, not claims about observed runtime usage. `preflight.sh` requires at least 600 MiB currently available before adding the storage pair. After deployment, record actual RSS before starting the evidence replicas; if the cap causes OOM/restarts, stop and resize rather than silently removing limits.

Port allocation

12379	Seaweed metadata-etcd client
12380	Seaweed metadata-etcd peer
19333	Seaweed master HTTP
19334	Seaweed master gRPC
18082	Seaweed volume HTTP
18083	Seaweed volume gRPC
18888	Seaweed filer HTTP
18889	Seaweed filer gRPC
18333	Seaweed S3 HTTP backend, loopback only
18334	Seaweed S3 gRPC, loopback/process-internal use
18443	evidence S3 TLS/immutability frontend on VPC address

All ports are explicit because SeaweedFS otherwise derives gRPC ports as HTTP port + 10000, and its normal volume default 8080 collides with the commissioned ChirpStack backend. Custom master/filer gRPC addresses must be expressed explicitly to SeaweedFS tools, for example `10.104.0.2:19333.19334` and

10.104.0.2:18888.18889; using only :19333/:18888 incorrectly derives 29333/28888. The preflight refuses any candidate port already in use before initial deployment.

Tracked vs protected files

Tracked:

```
compose.yml
release.env.example
hosts/*.env.example
filer.toml
s3.json.example
haproxy-evidence-s3.cfg.example
preflight.sh
```

Live protected inputs:

/etc/lorawan-cloud/seaweedfs/release.env	0600 root:root
/etc/lorawan-cloud/seaweedfs/host.env	0600 root:root
/etc/lorawan-cloud/seaweedfs/filer.toml	0644 root:root
/etc/lorawan-cloud/seaweedfs/s3.json	0640 root:1000
/etc/lorawan-pki/evidence-objectstore/ca.crt	0644 root:root
/etc/lorawan-pki/evidence-objectstore/<node>.*	0640 root:haproxy
/srv/seaweedfs/metadata-etcd	persistent
/srv/seaweedfs/master	persistent
/srv/seaweedfs/volume	persistent

Never commit live S3 access keys or the TLS private key.

S3 identities

Use unique role credentials, not one shared key:

```
gateway-evidence-ingest
  Read:lorawan-evidence
  List:lorawan-evidence
  Write:lorawan-evidence

gateway-mqtt-evidence-collector
  Read:lorawan-evidence
  List:lorawan-evidence
  Write:lorawan-evidence

gateway-evidence-verifier
  Read:lorawan-evidence
  List:lorawan-evidence
```

No runtime identity receives Admin. Create the lorawan-evidence bucket from a local privileged SeaweedFS administration session before installing the runtime-only identity file.

Commissioning order

Do not start all three nodes at once without checkpoints.

1. Pin SeaweedFS 4.41 + etcd 3.5.15 linux/amd64 image digests.
2. Run preflight on ulc-01, ulc-02, ulc-03. All must PASS.
3. Stage protected directories/configs on all three hosts.
4. Start swmeta-01 only; expect no quorum yet and make no filer claims.

5. Start swmeta-02; prove two-member quorum/health.
6. Start swmeta-03; prove 3 voters and common cluster ID.
7. Start SeaweedFS on ulc-01; verify only that node's process/listeners.
8. Start ulc-02 SeaweedFS; verify master membership and second rack/volume.
9. Start ulc-03 SeaweedFS; verify all three masters/racks/volumes/filers.
10. Create private bucket lorawan-evidence once.
11. Apply fs.configure for /buckets/lorawan-evidence/ with replication=010.
12. Prove the effective rule reports replication 010.
13. Issue three runtime S3 identities and install identical protected s3.json on all nodes.
14. Issue per-node internal TLS certs for evidence-objects.internal.lorawan.com.
15. Merge/validate the HAProxy method/header frontend one host at a time.
16. Configure evidence services to use the local :18443 TLS endpoint.
17. Run gateway-evidence-ingest objectstore-contract-write through HAProxy.
18. Verify the printed immutable fixture from another host using objectstore-contract-verify.
19. Record exact SHA/size/ref and retain the commissioning objects; do not delete them.

The dedicated one-Droplet-loss exercise remains Phase 15. Initial commissioning proves the configured two-copy placement, exact create-only behavior through the production endpoint, and cross-host readability without deliberately killing a healthy production node.

Application configuration

The evidence-service common environment is frozen to:

```
EVIDENCE_OBJECTSTORE_BACKEND=s3
EVIDENCE_S3_ENDPOINT=https://evidence-objects.internal.lorawan.com:18443
EVIDENCE_S3_REGION=us-east-1
EVIDENCE_S3_BUCKET=lorawan-evidence
EVIDENCE_S3_PREFIX=lorawan-gateway-evidence
EVIDENCE_S3_CA_FILE=/run/evidence/s3/ca.crt
EVIDENCE_S3_USE_PATH_STYLE=true
```

us-east-1 is used only as the SigV4 signing region expected by the S3-compatible endpoint; the physical SeaweedFS data center remains sgp1.

Live commissioning checkpoint — 2026-08-31

Infrastructure through the host HAProxy boundary is live PASS on all three nodes. SEAWEEDFS_CORE_3_NODE=PASS, SEAWEEDFS_REPLICATION_010_EMPIRICAL=PASS, EVIDENCE_OBJECTSTORE_TLS_BUNDLES=PASS, and HAProxy_EVIDENCE_S3_3_NODE=PASS are authoritative.

The retained replication fixture has fid 3,01e3ab96f3, size 89, SHA-256 bf981516163ff1e35d6315213458423860be84f0b7fe74269ac8d780577bb5b, and exactly two copies on distinct Droplet racks. The bucket is lorawan-evidence with replication 010 and volumeGrowthCount 2.

Active identical runtime s3.json SHA-256 is

310aa8b74145256bae9e15f759bacfc37d590a5b54c08c348c38ea7e0c6371f8, mode 0640 root:1000. Ingest/collector are read/list/write; verifier is read/list only. The earlier leaked credential set is retired permanently and must not be reused.

Object-store CA SHA-256 is c1dedc8cc6b58217e955cf763868b429dacdd933bbe7d9ffed147122e9d013fd. The production logical endpoint is https://evidence-objects.internal.lorawan.com:18443. HAProxy enforces TLS >=1.2, GET/HEAD, and PUT only with exactly one If-None-Match: *; other methods return 405 and invalid/unconditional PUT returns 428. Raw S3 remains loopback-only.

Full EVIDENCE_OBJECTSTORE=PASS is **not** yet claimed. The accepted production ingest binary is not currently staged on ulc-01/02/03, so S9 objectstore-contract-write plus cross-host objectstore-contract-verify remains the final application-contract gate.

Hard acceptance boundary

Do not write EVIDENCE_OBJECTSTORE=PASS until real servers prove all of:

```
3 Seaweed masters healthy
3 dedicated metadata-etcd voters healthy
3 volume servers present with three distinct rack labels
bucket rule = replication 010
raw S3 backend loopback-only
HAProxy TLS endpoint valid with internal CA
runtime endpoint rejects DELETE/POST/unconditional PUT
production Go helper: first create PASS
production Go helper: exact duplicate idempotent PASS
production Go helper: conflicting duplicate rejected PASS
production Go helper: concurrent create has exactly one winner PASS
exact bytes/SHA readable through another host's endpoint
```

Upstream compatibility statements are not enough; the production Go helper is the authority for the conditional-write contract used by this project.

Supporting runbook: 01-live-commissioning.md

This is the original project runbook. Check the current documentation baseline and actual host state before applying historical steps or changing a live system.

SeaweedFS Live Commissioning — One Boundary at a Time

This runbook is the live operator companion to `README.md`. It deliberately separates read-only discovery from image pulls, filesystem creation, metadata quorum bootstrap, SeaweedFS startup, bucket/identity creation, HAProxy rollout, and application acceptance.

Do not combine the phases. Stop at the first failed gate and preserve the output.

S0 — ulc-02 immutable-image + host preflight

Run this block **only on ulc-02** first. It is read-only with respect to running services and does not pull/start/restart a container.

```
sudo bash <<'SEAWEEED_S0'
set -Eeuo pipefail

fail() { echo "FAIL|$*" >&2; exit 1; }
pass() { echo "PASS|$*"; }

[ "$(hostname -s)" = "ulc-02" ] || fail "run S0 only on ulc-02"

SEAWEEED_TAG='chrislusf/seaweedfs:4.41'
ETCD_TAG='quay.io/coreos/etcd:v3.5.15'
CANDIDATE_PORTS='12379 12380 19333 19334 18082 18083 18888 18889 18333 18334 18443'

echo '=====
echo ' SEAWEEEDFS S0 - IMMUTABLE IMAGE + HOST PREFLIGHT'
echo ' READ ONLY / NO IMAGE PULL / NO CONTAINER START'
echo '=====

echo
echo '=== 1. HOST IDENTITY ==='
hostname -f || true
ip -4 -br addr
ip -4 addr show | grep -Fq '10.104.0.4/' || fail 'ulc-02 VPC address 10.104.0.4 missing'
pass 'ulc-02 VPC identity'

echo
echo '=== 2. DOCKER TOOLING ==='
docker --version
docker compose version
docker buildx version
pass 'Docker/Compose/Buildx available'

echo
echo '=== 3. RESOLVE IMMUTABLE REGISTRY DIGESTS ==='
resolve_ref() {
    local tag="$1"
    local label="$2"
    local out digest
```

```

out="$(docker buildx imagetools inspect "$tag")" || fail "$label registry inspection failed"
printf '%s\n' "$out" | grep -Fq 'linux/amd64' || fail "$label does not advertise linux/amd64"
digest="$(printf '%s\n' "$out" | awk '/^Digest:/ {print $2; exit}')"
[[ "$digest" =~ ^sha256:[0-9a-f]{64}$ ]] || fail "$label digest parse failed"

docker buildx imagetools inspect "${tag}@${digest}" >/dev/null || fail "$label digest lookup failed"
printf '%s|%s|%s\n' "$label" "$tag" "$digest"
}

SEAWEED_LINE="$(resolve_ref "$SEAWEED_TAG" SEAWEEDFS)"
ETCD_LINE="$(resolve_ref "$ETCD_TAG" SEAWEED_METADATA_ETCD)"
printf '%s\n' "$SEAWEED_LINE"
printf '%s\n' "$ETCD_LINE"
pass 'immutable registry references resolved and amd64 advertised'

echo
echo '=== 4. CURRENT MEMORY / DISK ==='
free -m
df -h / /srv 2>/dev/null || df -h /
MEM_AVAILABLE_KIB="$(awk '/MemAvailable:/ {print $2}' /proc/meminfo)"
[ "${MEM_AVAILABLE_KIB:-0}" -ge 614400 ] || fail 'less than 600 MiB MemAvailable'
DISK_TARGET=/srv
[ -d /srv ] || DISK_TARGET=/
DISK_AVAILABLE_KIB="$(df -Pk "$DISK_TARGET" | awk 'NR==2 {print $4}')"
[ "${DISK_AVAILABLE_KIB:-0}" -ge 5242880 ] || fail 'less than 5 GiB free on storage filesystem'
pass 'minimum current memory/disk headroom'

echo
echo '=== 5. EXISTING PATRONI DCS ETCD MUST REMAIN PRESENT ==='
ss -H -lnt 'sport = :2379' | tee /dev/stderr | grep -q . || fail '2379 listener missing'
ss -H -lnt 'sport = :2380' | tee /dev/stderr | grep -q . || fail '2380 listener missing'
pass 'existing etcd listeners remain present'

echo
echo '=== 6. SEAWEEDFS CANDIDATE PORTS MUST ALL BE FREE ==='
for port in $CANDIDATE_PORTS; do
    if ss -H -lnt "sport = :$port" | grep -q .; then
        ss -lntp "sport = :$port" || true
        fail "candidate port already in use: $port"
    fi
    echo "port_${port}=FREE"
done
pass 'all SeaweedFS candidate ports free on ulc-02'

echo
echo '=== 7. EXISTING CONTAINER RESOURCE SNAPSHOT ==='
docker ps --format 'table {{.Names}}\t{{.Status}}\t{{.Ports}}'
echo
docker stats --no-stream --format 'table {{.Name}}\t{{.CPUPerc}}\t{{.MemUsage}}' || true

echo
echo '===== '
echo ' S0 RESULT '
echo '===== '
echo 'SEAWEEDFS_S0_ULC02=PASS'
echo 'NO_IMAGE_PULLED_BY_THIS_BLOCK=YES'
echo 'NO_CONTAINER_STARTED_BY_THIS_BLOCK=YES'
echo 'NEXT=PIN_THE_TWO_PRINTED_IMAGE_REFS_THEN_PULL_INSPECT_ON_ULC02'
SEAWEED_S0

```

Expected output includes exactly two immutable reference lines:

```

SEAWEEDFS|chrislusf/seaweedfs:4.41@sha256:<64-hex>
SEAWEED_METADATA_ETCD|quay.io/coreos/etcd:v3.5.15@sha256:<64-hex>

```

Do not copy a digest from documentation or another architecture. The registry result from the approved Docker/Buildx host is the input for the next boundary.

Live boundary status — 2026-08-31

```
S0 PASS - immutable-image/host discovery completed
S1 PASS - SeaweedFS 4.41 and metadata-etcd immutable refs inspected
S2 PASS - three-host resource/port preflight completed
S3 PASS - protected SeaweedFS directories/config staged
S4 PASS - swmeta-01/02/03 form a healthy 3-voter metadata-etcd quorum
S5 PASS - SeaweedFS core healthy on ulc-01/02/03
S6 PASS - lorawan-evidence bucket + replication=010; empirical two-rack placement proven
S7 PASS - least-privilege runtime S3 identities + internal object-store PKI installed
S8 PASS - HAProxy TLS/create-only boundary proven on all three nodes
S9 PENDING - production gateway-evidence-ingest helper binary is not yet staged on any server
S10 BLOCKED ON S9 - do not claim EVIDENCE_OBJECTSTORE=PASS yet
```

Pinned live images:

```
SeaweedFS  chrislusf/seaweedfs:4.41@sha256:43b768cd62b00d132439cda881b93fd1adebf1b315e996e794087743821d771d
metadata
quay.io/coreos/etcd:v3.5.15@sha256:0934690612905554eb61ddefb9faaaecb47c2f6931dbb453e694358092ee8990
```

Live custom ports are master 19333/19334, volume 18082/18083, filer 18888/18889, raw S3 127.0.0.1:18333, S3 gRPC 18334, and HAProxy TLS 18443. When using SeaweedFS CLI/shell with the custom master/filer gRPC ports, use explicit combined address forms such as 10.104.0.2:19333.19334 and 10.104.0.2:18888.18889; plain :19333 or :18888 incorrectly derives 29333 or 28888.

The one-Droplet-loss kill/recovery exercise remains Phase 15 and is not part of this streamlined initial commissioning.

Supporting runbook: 06-replicated-ha-deployment-journey.md

This is the original project runbook. Check the current documentation baseline and actual host state before applying historical steps or changing a live system.

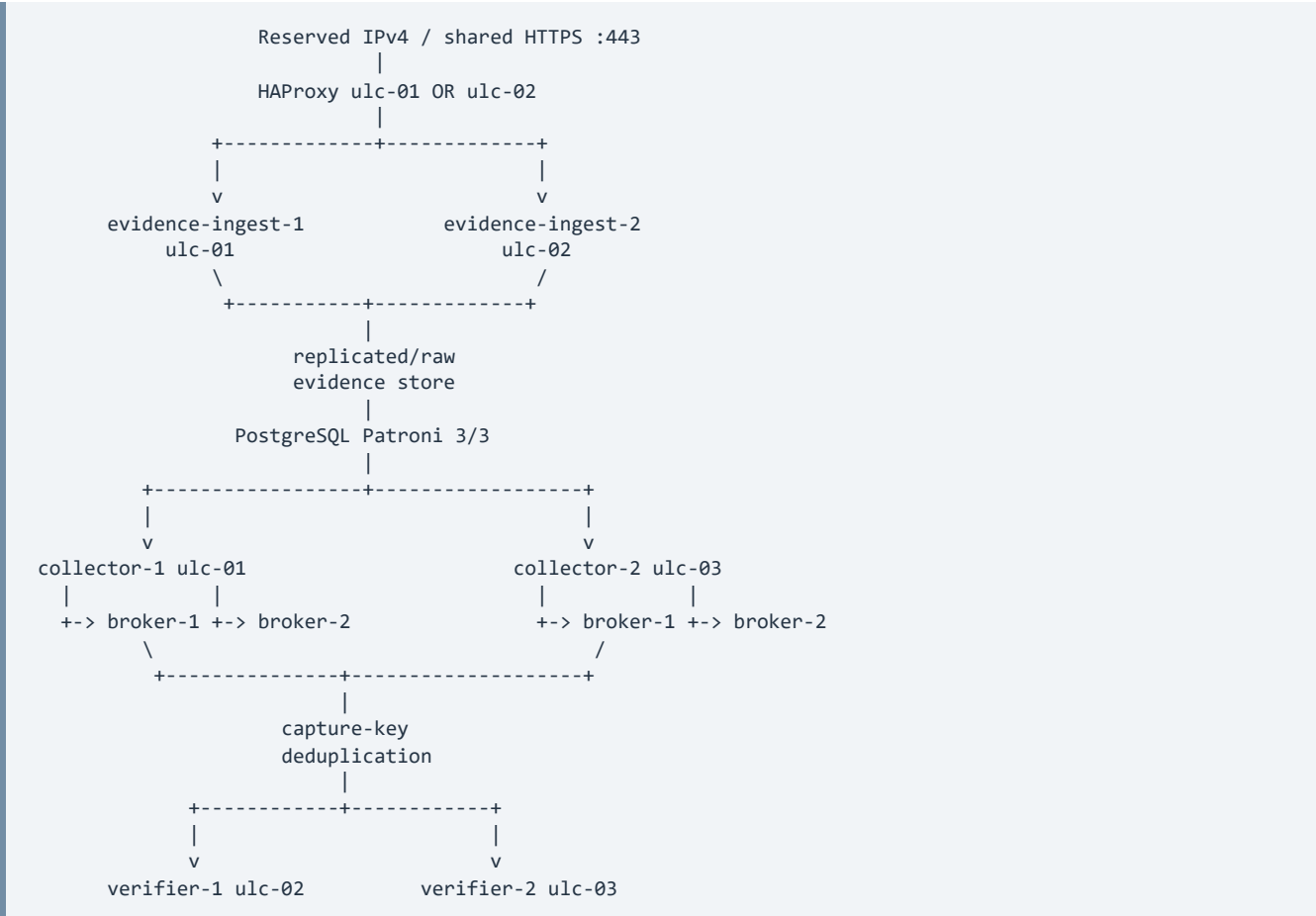
Replicated Gateway Evidence Services - Streamlined Deployment Journey

Status: LIVE COMMISSIONING IN PROGRESS. Source/build readiness, SeaweedFS infrastructure through the HAProxy boundary, the reviewed gateway_evidence migration, and the three-node evidence-aware PostgreSQL HBA are already live PASS. Production object-store helper S9, evidence workload LOGIN/PgBouncer closure, PKI/ingress, and service replicas remain. Preserve completed gates and resume only from the current failed boundary.

6.1 Goal

Build the evidence tier so one cloud service instance can disappear without losing an accepted evidence object or creating two authoritative verification results.

Target runtime shape after capacity validation:





The gateway journal/uploader remain single-device services because there is one physical gateway. Their resilience comes from crash-safe local state, independent MQTT delivery buffering, and cloud checkpoints; do not call them an HA pair.

6.2 Non-negotiable HA semantics

```

Ingest
  active/active
  same retry -> one accepted identity
  same identity + different digest -> security conflict

Collector
  active/active
  two replicas
  each replica -> persistent read-only session to broker-1 AND broker-2
  duplicate observations -> one deterministic capture_key_sha256

Verifier
  active/active
  two replicas
  FOR UPDATE SKIP LOCKED + expiring lease
  crash -> lease expires -> other worker reclaims

Trusted decoder
  stateless
  same immutable code/image digest on both verifier replicas

Raw storage
  one-Droplet loss cannot remove the only copy
  create-only/no-overwrite
  exact-byte SHA-256 recovery proof required

Database
  existing 3-node Patroni cluster is the authoritative metadata/state layer
  
```

6.3 Why the setup is split into guarded blocks

One giant script that mutates storage, PostgreSQL, HAProxy, PKI, three services, and Grafana makes failure recovery ambiguous. Instead, use one block per trust boundary. Keep initial commissioning deliberately small: prove each replica starts, then run one representative convergence/function check for the pair. Deep failure testing stays in Guide 3 / Phase 15.

Standard wrapper behavior:

```

preflight -> install/start replica 1 -> health -> install/start replica 2 -> health -> one
functional/convergence smoke -> evidence -> PASS
  
```

If a block fails after the canary passes, preserve that PASS and resume from the failed replica/check. Do not rerun the whole phase automatically.

Standard guarded one-block shell pattern

Every future live block should follow the same controlled child-shell shape below. This is a **structural template**, not a command to run now: a delivered live block must replace the marked body with reviewed, concrete commands and remove the template stop gate before the operator executes it.

```
sudo bash <<'EOF'
set -euo pipefail

# TEMPLATE SAFETY: a real reviewed block removes this guard.
echo 'TEMPLATE_ONLY=STOP'
exit 64

STEP='<BOUNDARY_NAME>'
RUN_ID="EVIDENCE-SVC-$(date -u +%Y%m%dT%H%M%SZ)"
EVIDENCE_DIR="/home/opsadmin/lorawan-ha-evidence/${RUN_ID}/<STEP_DIR>"
install -d -m 700 "$EVIDENCE_DIR"

fail() {
    printf 'STEP=%s\nRESULT=FAIL\nREASON=%s\n' "$STEP" "$*" >&2
    exit 1
}

pass() {
    printf '%s=PASS\n' "$1"
}

# -----
# 1. READ-ONLY PREFLIGHT
#   - previous PASS prerequisites
#   - current service/cluster/resource state
#   - exact target files/listeners free/expected
# -----

# -----
# 2. CANARY MUTATION
#   - minimum rollback copy for this boundary only
#   - validate candidate before reload/start
# -----

# -----
# 3. VERIFY CANARY
#   - health/restart/OOM
#   - permissions/ACLs
#   - functional positive + negative checks
# -----
pass REPLICAS_1

# -----
# 4. SECOND REPLICAS MUTATION
#   - execute only after REPLICAS_1 passed
# -----

# -----
# 5. VERIFY SECOND REPLICAS
# -----
pass REPLICAS_2

# -----
# 6. HA INVARIANT
#   - duplicate work converges safely
#   - durable state is shared/replicated as designed
#   - no unrelated service regression
# -----
pass HA_INVARIANT

# -----
# 7. SECRET-FREE EVIDENCE
#   - bounded command output/config hashes only
#   - never copy .env/private keys/passwords/tokens/SecretIDs
# -----
find "$EVIDENCE_DIR" -type d -exec chmod 700 {} +
```

```

find "$EVIDENCE_DIR" -type f -exec chmod 600 {} +
(
  cd "$EVIDENCE_DIR"
  find . -type f ! -name SHA256SUMS -print0 \
    | sort -z \
    | xargs -0 -r sha256sum > SHA256SUMS
  sha256sum -c SHA256SUMS
)

printf 'STEP=%s\n' "$STEP"
printf 'UTC=%s\n' "$(date -u +%Y-%m-%dT%H:%M:%SZ)"
printf 'REPLICA_1=PASS\n'
printf 'REPLICA_2=PASS\n'
printf 'HA_INVARIANT=PASS\n'
printf 'EVIDENCE_PATH=%s\n' "$EVIDENCE_DIR"
printf '<BOUNDARY_PASS_MARKER>=PASS\n'
EOF

```

Why this pattern is preferred:

- it keeps one logical deployment boundary in one copy/paste operation;
- replica 2 is unreachable if the canary fails because the child shell exits immediately;
- successful early gates can be preserved if a later verification fails;
- evidence generation is part of the block instead of a separate forgotten step;
- the block does not create another full backup cycle;
- the top-level login shell is not left in set -euo pipefail mode;
- future wrappers that call SSH must use ssh -n for no-input checks or redirect from an explicit payload file so SSH cannot consume the parent heredoc.

For each actual boundary in Sections 6.5-6.12, the project session should produce the concrete reviewed version of this block only when all required image names, paths, identities, ports, and rollback actions are known.

6.4 Journey evidence root

For live deployment use one protected run tree on the control host:

```

/home/opsadmin/lorawan-ha-evidence/EVIDENCE-SVC-<UTC>/
00-preflight/
01-storage-db/
02-pki-ingress/
03-ingest/
04-collector/
05-verifier-decoder/
06-observability/
07-normal-path/
SHA256SUMS

```

Directory mode 0700; evidence files 0600. Do not copy .env, private keys, database passwords, OpenBao tokens, SecretIDs, recovery material, or Fabric client keys into this tree.

6.5 Block 0 - artifact and capacity preflight

Mutation: none.

Require before any live evidence service is created:

```

reviewed source revision for all three server services
immutable image/executable digest for each
runtime UID/GID

```

```
health/readiness contract
configuration schema
trusted-decoder immutable digest + fixtures
v2 canonicalization vector
selected replicated evidence-storage technology
reviewed gateway_evidence migration/grants
reviewed shared-443 design
collector TLS/client-ID design for both brokers
rollback commands
```

The one-block preflight must inspect all three hosts for memory, disk, listeners, current containers, restart/OOM state, and the exact ports/mounts the candidate would consume. Do not start anything.

Pass marker:

```
EVIDENCE_REPLICATED_PREFLIGHT=PASS
```

6.6 Block 1 - replicated evidence storage + PostgreSQL migration

Mutation: storage namespace/backend plus one reviewed PostgreSQL migration only.

Guarded order inside one block:

1. re-prove Patroni 1 leader / 2 replicas and selected durable storage healthy
2. create the protected evidence namespace/bucket/path with create-if-absent policy
3. write one synthetic non-secret object and verify its exact SHA-256
4. apply the reviewed versioned `gateway\_evidence` migration once on the current primary
5. verify tables/constraints/indexes/lease columns/capture-key uniqueness and grants
6. verify both PostgreSQL replicas received the schema
7. delete only the synthetic storage probe if the backend contract permits test cleanup
8. capture hashes/metadata and PASS

Do not run a destructive storage failure here; Phase 15 owns actual member-loss testing.

Pass markers:

```
EVIDENCE_STORAGE_REPLICATION=PASS
EVIDENCE_DB_MIGRATION=PASS
```

6.7 Block 2 - evidence PKI + shared HTTPS ingress

Mutation: evidence certificates/identities and reviewed HAProxy shared-443 configuration only.

Canary-first order:

1. verify current ChirpStack HTTPS before change
2. verify anchor ownership/listeners
3. issue/stage evidence server certificate and per-gateway clientAuth identity without printing private material
4. validate SAN/EKU/chain/fingerprint/expiry/EUI mapping
5. build HAProxy candidate for ulc-01
6. haproxy -c before reload
7. activate canary
8. prove ChirpStack browser/API HTTPS still works without a client cert
9. prove evidence SNI route rejects no-cert/unknown-client
10. prove valid evidence client reaches only evidence backend
11. only then repeat the reviewed config on ulc-02
12. prove both anchor candidates equivalent
13. capture config hash/cert metadata and PASS

The exact SNI/client-cert method remains blocked until the reviewed HAProxy 2.8 configuration exists. Do not globally require a gateway certificate on ordinary ChirpStack HTTPS.

Pass marker:

```
EVIDENCE_PKI_INGRESS=PASS
```

6.8 Block 3 - two active/active ingest replicas

Mutation: deploy ingest-1 then ingest-2.

Frozen target placement:

```
ulc-01 -> evidence-ingest-1
ulc-02 -> evidence-ingest-2
```

One block may do both replicas only with this gate sequence:

1. verify storage + DB + PKI/ingress PASS markers and stage the identical immutable image/config on both hosts
2. install distinct ingest-1 credential; start ingest-1; verify health/runtime UID/listener/resource state
3. install distinct ingest-2 credential; start ingest-2; verify the same health gates
4. send one approved synthetic checkpoint/segment fixture through the normal ingress/backend path
5. repeat the exact fixture once and prove one raw object + one metadata identity remain
6. prove object SHA matches metadata and both replicas remain healthy
7. capture evidence and PASS

Pass marker:

```
EVIDENCE_INGEST_REPLICATED=PASS
```

6.9 Block 4 - two active/active MQTT evidence collectors

Mutation: dedicated collector identities/ACLs and collector-1/2 runtime.

Frozen target placement:

```
ulc-01 -> collector-1
ulc-03 -> collector-2
```

Each collector has **two broker sessions**:

```
collector-N/broker-1 -> 10.104.0.2:8884, TLS identity mqtt.internal.lorawan.com
collector-N/broker-2 -> 10.104.0.4:8884, TLS identity mqtt.internal.lorawan.com
```

Use distinct client IDs for all four sessions and persistent subscriptions. ACL is read-only as923/gateway/+/event/#; publish and command topics are denied.

Guarded order:

1. verify both broker backends, TLS identity, and the reviewed read-only ACL configuration
2. stage/start collector-1 and prove its two persistent broker sessions
3. stage/start collector-2 and prove its two persistent broker sessions
4. feed one approved non-RF commissioning fixture through the reviewed safe input path when available
5. prove duplicate observations converge to one `capture\_key\_sha256` and one logical raw-object identity

```
6. capture the four client/session health states without secrets
7. PASS
```

Do not interpret the duplicate observation by collector-1/2 as two gateway events.

Pass marker:

```
EVIDENCE_MQTT_COLLECTOR_REPLICATED=PASS
```

6.10 Block 5 - two verifier workers + trusted decoder

Mutation: verifier-1/2 runtime only; no OpenBao/Fabric permission.

Frozen target placement:

```
ulc-02 -> verifier-1 + trusted decoder
ulc-03 -> verifier-2 + trusted decoder
```

Both must report the exact same decoder digest/version before either can process work.

Guarded order:

```
1. prove one fixed valid decoder vector on both replicas and require identical decoder digest/version
2. start verifier-1 and verify health + least-privilege DB configuration
3. start verifier-2 and verify the same health/configuration gates
4. queue one fixed pending verification fixture
5. prove exactly one worker owns/processes it and the terminal state/reason/digests match the fixture
6. prove no duplicate terminal row/work item appeared and both replicas remain healthy
7. PASS
```

Actual host/process failure timing, lease-expiry/reclaim, SKIP LOCKED distribution under concurrent load, and tamper/reorder/delete/checkpoint-conflict fixtures belong to Guide 3 / Phase 15. They are not required to get the replicated verifier pair installed and working.

Pass markers:

```
EVIDENCE_VERIFIER_REPLICATED=PASS
TRUSTED_DECODER=PASS
```

6.11 Block 6 - read-only evidence observability

Mutation: Grafana provisioning/read-only views only.

Add panels/queries for:

```
replica health ingest-1/2
collector-1/2 broker-session health
verifier-1/2 worker/lease state
latest checkpoint age per gateway
pending verification age
verified/evidence_gap/integrity_failure counts
collector unmatched count
trusted-decoder mismatch count
raw-store replication/member health
v2 outbox rows blocked on verification
```

Telemetry freshness and evidence freshness remain different panels.

Pass marker:

```
EVIDENCE_OBSERVABILITY=PASS
```

6.12 Block 7 - real normal-path lineage

This step waits for the physical gateway and reviewed server runtimes.

One real staging uplink must produce exactly one lineage:

```
journal record
-> closed segment/checkpoint
-> replicated evidence ingest
-> one logical MQTT capture despite collector replication
-> accepted ChirpStack application event
-> trusted decoder result
-> TimescaleDB row
-> one event_verification row = verified
```

Then and only then may the matching v2 outbox row become eligible for the Fabric Adapter.

Pass marker:

```
GATEWAY_EVIDENCE_V2_NORMAL_PATH=PASS
```

6.13 What gets written to Markdown after every block

After every operator block, record only verified facts:

```
UTC/date
boundary name
hosts/replicas touched
immutable image/executable hashes
config/migration hash where relevant
non-secret certificate metadata where relevant
resource snapshot
named PASS markers
any harness defect and exact resume point
what was deliberately not repeated
next boundary
```

Update:

```
deployment/server/integrations/gateway-integrity/<relevant-guide>.md
deployment/server/cloud-production/00-build-execution-log.md
deployment/server/cloud-production/00-current-server-continuation-checkpoint.md
```

If a block partially passes, write the successful gates immediately and resume from the first failed gate. Do not discard valid evidence and do not rerun successful replicas only for ceremony.

6.13A Live execution checkpoint — 2026-08-31

This section is the authoritative resume point after the interrupted documentation period. Do not rerun completed boundaries merely to reconstruct evidence.

Raw evidence infrastructure

```
SeaweedFS release      4.41 / commit de34a1a87
SeaweedFS image
chrislusf/seaweedfs:4.41@sha256:43b768cd62b00d132439cda881b93fd1adebf1b315e996e794087743821d771d
metadata-etcd image
quay.io/coreos/etcd:v3.5.15@sha256:0934690612905554eb61ddefb9faaaecb47c2f6931dbb453e694358092ee8990
metadata-etcd ports    12379 / 12380
master                 19333 HTTP / 19334 gRPC
volume                 18082 HTTP / 18083 gRPC
filer                  18888 HTTP / 18889 gRPC
raw S3                  127.0.0.1:18333
S3 TLS boundary        node VPC :18443
replication             010 = one additional copy, different rack, same sgp1 DC
```

SEAWEEDFS_CORE_3_NODE=PASS and SEAWEEDFS_REPLICATION_010_EMPIRICAL=PASS are authoritative. The retained placement fixture is 3,01e3ab96f3, size 89, SHA-256 bf981516163ff1e35d6315213458423860be84f0b7fe74269ac8d780577bb5b, with copies on two distinct Droplet racks. The lorawan-evidence bucket is configured with replication 010 and volume growth count 2.

Runtime S3 identities are installed for ingest, collector, and verifier. Ingest/collector have read/list/write; verifier has read/list only. The active s3.json is identical on all three hosts, SHA-256 310aa8b74145256bae9e15f759bacfc37d590a5b54c08c348c38ea7e0c6371f8, protected as 0640 root:1000. The earlier leaked S3 credentials were retired and must never be reused.

Internal object-store PKI is live. CA SHA-256 is c1dedc8cc6b58217e955cf763868b429dacdd933bbe7d9ffed147122e9d013fd; logical endpoint is evidence-objects.internal.lorawan.com:18443. HAProxy on ulc-01/02/03 accepts GET/HEAD and only conditional PUT with exactly one If-None-Match: *; other methods are 405 and bad/unconditional PUT is 428. Raw Seaweed S3 remains loopback-only. HAProxy_EVIDENCE_S3_3_NODE=PASS is authoritative. Retained HAProxy fixtures exist on all three nodes and must not be deleted merely to tidy commissioning evidence.

S9 COMPLETE / PASS on 2026-09-01. The locked production gateway-evidence-ingest helper executed objectstore-contract-write through https://evidence-objects.internal.lorawan.com:18443; the retained object and race fixture were then verified cross-host from ulc-03. Record OBJECTSTORE_CONTRACT=PASS, SEAWEEDFS_S9_WRITE=PASS, SEAWEEDFS_S9_CROSS_HOST_VERIFY=PASS, and EVIDENCE_OBJECTSTORE=PASS. Retain the commissioning fixtures and do not repeat S0-S9 without a storage change or failure.

gateway_evidence database migration

The live migration is COMPLETE / PASS. Repository source hashes remain:

```
001_gateway_evidence.sql      bf2a1e3188cf67107872c425064d55fb476d7ea58855510b154f7e869795a8b9
001_gateway_evidence.verify.sql e08112fd48cdb6be058f40487ac7fff4d4b60a9647f2e1750cf1ebbd974ed4ae
```

The reviewed terminal transport copies that were actually executed had hashes cd03a10f39dc4e780be5f7c7718596816a199dfaf118d7bd9c3f1c5ce4e1a630 and fc57361eda4fc7fb20e3148eed421e2f653f38dfc25e00e6dee07dba59adc040; preserve that distinction instead of claiming byte identity with the repository files.

Pre-migration backup on current leader ulc-01 is /root/backups/evidence-db-pre-migration/20260831T074754Z/lorawan_telemetry.dump, SHA-256

f9564ffaf8e65e021fa025cd45e70c9472608bfa7b6e4912cc568cc773aa21de, size 79962, with pg_restore --list PASS. The Timescale continuous_agg circular-FK warning was informational because this was a full custom-format dump, not a data-only dump.

EVIDENCE_DB_MIGRATION=PASS, EVIDENCE_DB_PRIMARY_VERIFY=PASS, and EVIDENCE_DB_3_NODE_REPLICATION=PASS are authoritative. The migration created four gateway_evidence base tables, three nullable telemetry provenance columns, six relevant indexes, the monotonic checkpoint trigger, and three passwordless NOLOGIN authority roles:

```
gateway_evidence_ingestor
gateway_evidence_collector
gateway_evidence_verifier
```

Evidence PostgreSQL HBA rollout

The HBA extension is COMPLETE / PASS on all three Patroni members. The policy remains exactly 20 rules: no broad rule was added. Instead, the existing three hostssl lorawan_telemetry ... /32 scram-sha-256 rules now use this user selector:

```
telemetry_writer,telemetry_reader,fabric_adapter,+gateway_evidence_ingestor,+gateway_evidence_collector,+gateway_evidence_verifier
```

All three live/persistent HBA lists have SHA-256

a943358a884249aaae74b663a81fa6dde2d7c98deeb31f93def8e5bb4aa729f1, zero parse errors, and the two final reject rules remain intact. Rollout order was ulc-03 canary/persist -> ulc-02 canary/persist -> ulc-01 leader canary/fresh-auth/persist. No Spilo restart/recreate or Patroni switchover occurred.

Three harness defects were corrected during the rollout and must not be reintroduced:

1. Patroni reload is asynchronous; poll pg_hba_file_rules for convergence instead of sleep 3.
2. OLD_USERS is a prefix of NEW_USERS; compare the parsed HBA user field exactly, not substring containment.
3. pg_stat_replication.client_addr is inet; use host(client_addr), not client_addr::text, for exact address matching.

Before leader persistence, brand-new physical-replication IDENTIFY_SYSTEM sessions were created from ulc-02 source 10.104.0.4 and ulc-03 source 10.104.0.8 using standby, SCRAM, and sslmode=verify-full. Both returned system identifier 7676855802088521796, timeline 3, LSN 0/68000000. FRESH_REPLICATION_AUTH_TO_ULC01=PASS is authoritative.

Evidence database credentials + PgBouncer rollout

The planned identities are:

```
evidence_ingest_ulc01    -> gateway_evidence_ingestor
evidence_ingest_ulc02    -> gateway_evidence_ingestor
evidence_collector_ulc01 -> gateway_evidence_collector
evidence_collector_ulc03 -> gateway_evidence_collector
evidence_verifier_ulc02  -> gateway_evidence_verifier
evidence_verifier_ulc03  -> gateway_evidence_verifier
```

Historical issuance note: the first evidence_ingest_ulc01 login exposed the missing group-level database CONNECT grant, producing permission denied for database "lorawan_telemetry". That stop was correctly attributed to the database ACL layer rather than HBA/TLS/SCRAM and was fixed without rotating the existing credential. The PASS results below supersede that intermediate stop.

CONNECT authority correction: PASS on 2026-09-01. After two harness-only attempts that stopped before mutation, the corrected gate re-proved pre-state t|t|t|t|f|f|f|f, granted CONNECT ON DATABASE

lorawan_telemetry only to gateway_evidence_ingestor, gateway_evidence_collector, and gateway_evidence_verifier, and proved post-state `t|t|t|t|t|t|t|t` on all three PostgreSQL members while `PUBLIC` remained denied. The existing `evidence_ingest_ulc01` password was not regenerated; its fresh direct SCRAM + verify-full session succeeded and its exact least-privilege role structure remained intact.

Evidence LOGIN issuance: COMPLETE / PASS on 2026-09-01. Safe-resume preserved `evidence_ingest_ulc01`, created only the five absent roles, and closed all six with unique protected SCRAM credentials, one exact intended NOLOGIN authority membership, inherited database `CONNECT`, direct verify-full authentication, and three-node catalog state `6|6|6|6|6|6`. PgBouncer remained unchanged at four entries with zero evidence identities throughout.

PgBouncer three-node evidence expansion: COMPLETE / PASS on 2026-09-01. The final live inspection found `ulc-01`, `ulc-02`, and `ulc-03` already converged on the same ten-entry/six-evidence SCRAM userlist, mode `0640` root:postgres, size 1582 bytes, SHA-256 `665f1592c96ca276681a454b9cbcd6ab8ab0cbfb4594b8ddd443239db58df391`. The prior four-role file is SHA-256 `e159c1c3e408eb091629083a81738c730bb8d2ab3c07ecbc6aaf109c51b9d5e8`. Rollback directories are `/etc/pgbouncer/evidence-userlist-rollout-20260901T004328Z` on `ulc-01`, `...T020032Z` on `ulc-02`, and `...T020244Z` on `ulc-03`. The `ulc-02/ulc-03` rollback sets prove byte-identical before/candidate verifiers for all four pre-existing roles. PgBouncer remains active/enabled with PIDs 1698991, 1512906, and 1690020, matching their pre-expansion recorded PIDs and therefore proving reload without process restart.

A fresh strict verify-full acceptance used three commissioned evidence roles against physical endpoints `10.104.0.2:6432`, `.4:6432`, and `.8:6432`. All three authenticated, negotiated TLS 1.3 / TLS_AES_256_GCM_SHA384, and reached writable `lorawan_telemetry` on current Patroni primary `10.104.0.2`. PgBouncer expansion is no longer a resume item. SeaweedFS S9, PKI/collector identities, immutable OCI rollout, replicated service readiness, shared-443, and Grafana evidence views subsequently passed; the cloud journey is now `EVIDENCE_CLOUD_RUNTIME=PASS`.

6.14 Final commissioning rule

```
replicated process != replicated truth
```

HA is accepted only when durable state makes duplicate execution safe and one-instance loss recoverable:

Ingest replicas	-> idempotent stable identities
Collector replicas	-> deterministic capture key
Verifier replicas	-> durable leases
Trusted decoder replicas	-> identical immutable code
Raw evidence store	-> cross-host durable bytes
PostgreSQL	-> Patroni HA
OpenBao	-> Raft HA
Fabric adapter	-> lease-safe dual workers

No component may claim replication merely because `replicas: 2` appears in Compose.

6.15 Current cloud commissioning result - 2026-09-01

The complete server-side journey is PASS: raw-object storage S0-S9, database roles/PgBouncer, Evidence PKI, collector mTLS identities/ACLs, replicated evidence services, disabled Fabric standbys, shared-443 mTLS passthrough, MQTT witness capture, and read-only Grafana checkpoint/verification panels. The subsequent Gateway-01 target-package and real assembled EMU-01 physical-lineage gate is also PASS; preserve this normal path and move remaining work to sensor test execution, provider failover acceptance, or deliberate Fabric activation.